CatchUp AI · system architecture (as implemented, Nov 2026)

Dashed amber lines are trust boundaries. Numbers show one recovery request. Every box exists in the code; nothing here is planned.

STUDENT DEVICE (UNTRUSTED)

Nature Flow 3.0 web app
Next.js 16 / React 19 pages
Mission control · Copilot · Recovery lab · Schedule · Universe (3D + list)
Light / dark / system · keyboard
PWA install

iPhone app (SwiftUI)
Native shell around the same app

Student decisions
Upload notes · ask the copilot
Review exact changes
Approve or reject

MCP apps (optional)
Personal revocable token, read-only tools

VERCEL · SERVER FUNCTIONS IN SEOUL (icn1) · TRUSTED

Edge of the app: proxy and route handlers
Per-request CSP nonce · Origin check on changes · verified session (ES256 JWT) · Zod validation
Per-student rate limits · AI gate: kill switch, daily token budget, one run at a time · request ids, redacted logs

LearningRecoveryAgent
OpenAI Agents SDK runner, ≤ 8 turns, 90 s

The model decides:
which tools, in what order, with what arguments; what to search; which study items to propose; when to ask instead

The server decides:
identity, permissions, budgets, and which facts survive (claim check before display)

Structured answer: facts · suggestions

7 tools (owner re-checked)
get_study_preferences · read
list_assignments · read (titles flagged)
search_course_material · read, cited
get_learning_progress · read
get_student_memory · read
suggest_memory · suggestion only
draft_recovery_plan · PENDING draft only

No tool can save, approve or delete.

RAG over the student's PDFs
Page-level chunks · keyword + vector (RRF)
Abstains when nothing supports an answer
Flags text that tries to instruct the AI
Claim-to-citation check on every fact
Signed 120 s links to the exact page

Deterministic planner (not AI)
Free time per local day (time zone, DST)
Earliest deadline first · blocks ≤ 90 min
Three routes: deadline-first, balanced, full
Feasible or short by N minutes
Same inputs → same plan

Approval endpoint → approve_proposal (database transaction)
Row lock · bound to the exact draft and its input hash · refuses stale, expired or infeasible drafts
Idempotency key: 10 parallel approvals → exactly 1 plan · writes the study schedule
Every run's tool calls, timings, model and tokens are stored for the "Why trust this?" panel

Schedule, focus sessions, quizzes, insights, recovery score
Read back under the student's own permissions · calendar export (private, revocable feed)
/quality page and /api/health/ready for judges and monitoring

SUPABASE · SEOUL · TRUSTED DATA

Auth
Email code / magic link · Turnstile bot check
Anonymous demo accounts (deleted after 24 h)
ES256-signed sessions

PostgreSQL
Row-level security on all 25 tables
Composite owner keys · column grants
pgvector + full-text search function
Proposals, plans, tasks, agent runs
Rate limits · ops flags (kill switch)
Memory (student-confirmed only)
Migrations 0001–0015, applied

Private storage
PDF only, 10 MB, per-student folders
Validated before indexing; deletion removes file, chunks and search hits

↓ **agent model + embeddings (server-side key)**

EXTERNAL SERVICES

Google Gemini API
Flash model + gemini-embedding (key on server)

GitHub Actions
CI on every push
Uptime every 30 min
Manual live eval

Mail + bot check
Gmail SMTP codes
Opt-in reminders
Cloudflare
Turnstile

1 student asks · **2** agent picks tools · **3** tools read only that student's rows · **4** cited passages · **5** planner computes routes, draft saved as PENDING · **6** student reviews and approves · **7** one locked transaction commits · **8** schedule and insights